Seminar 1 - Linear Algebra

ECE 447: Data Analysis and Machine Learning for Engineers

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Land Acknowledgment

The University of Alberta acknowledges that we are located on Treaty 6 territory, and respects the histories, languages, and cultures of First Nations, Métis, Inuit, and all First Peoples of Canada, whose presence continues to enrich our vibrant community.

Physics Laureate Kip Thorne on Mentoring the Next Generation

"My mentoring has been one of my greatest joys. I am proud that I have mentored fifty something, fifty-two or fifty-four, I do not even know the number, of PhD students during my career and that they have gone in a huge number of different directions because I mentored them broadly so that they had the tools to be able to work in anything from an analyst at the CIA at one extreme, to very abstract theoretical physics, at the another extreme - in the computer industry, in management. I had two students who moved into the financial world very, very successfully.

"They look back and they say far more useful to them than Harvard Business School, which was a key piece for what they wanted to do, was doing a theoretical physics PhD under me. Because in doing a theoretical physics PhD they learned how to take a complex problem, break it down into pieces that could be solved, how to transform a problem into a soluble form, and that general skill that is the essence of success in physics, is transferable into all these different areas of human enterprise. So yes, I am proud of my mentoring and I have taken great joy in it."

Kip Thorne made crucial contributions to the development of the LIGO detector. In this detector, laser technology is used to measure small changes in length caused by gravitational waves. In 2015, gravitational waves were detected for the first time and in 2017 he was awarded the Nobel Prize in Physics.

Credit: Nobel Prize Facebook page

Nobel Prize
11h ·

This Seminar Covers ...

Basic of Linear Algebra

- Column space
- Nullspace
- Rank
- Vector spaces
- Subspaces
- Linear independence
- Basis
- Span, ...

We Deal with **Matrices and Vectors** in ECE 447

Bold lower-case for vectors, and bold upper-case for matrices

$$\boldsymbol{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^n, \qquad \boldsymbol{X} = \begin{bmatrix} - \boldsymbol{x}_1^\top - \\ - \boldsymbol{x}_2^\top - \\ \vdots \\ - \boldsymbol{x}_n^\top - \end{bmatrix} \in \mathbb{R}^{n \times p}, \qquad \varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{bmatrix} \in \mathbb{R}^n.$$

We will see equations like this

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\theta} + \varepsilon$$

Column Vectors and Linear Combinations

Consider the matrix

$$\boldsymbol{A} = \begin{bmatrix} | & | & & | \\ \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_p \\ | & | & & | \end{bmatrix} \in \mathbb{R}^{n \times p}$$

The vector $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}$ is a linear combination of column vector $\boldsymbol{v}_i$ weighted by x_i.

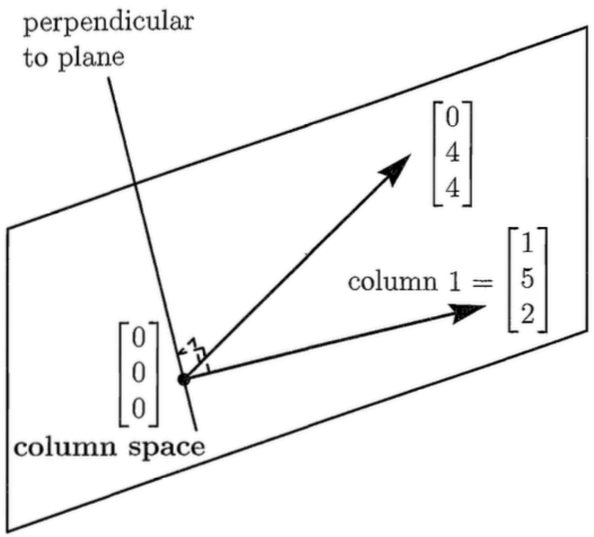
$$\boldsymbol{y} = x_1 \begin{bmatrix} | \\ \boldsymbol{v}_1 \\ | \end{bmatrix} + x_2 \begin{bmatrix} | \\ \boldsymbol{v}_2 \\ | \end{bmatrix} + \cdots + x_n \begin{bmatrix} | \\ \boldsymbol{v}_p \\ | \end{bmatrix}$$

What are the properties of matrix $\boldsymbol{A}$?

The Column Space of $\boldsymbol{A}$

Column Space $\boldsymbol{C}(\boldsymbol{A})$

Contains all linear combinations of the columns of $\boldsymbol{A}$.



Let $\boldsymbol{x}$ be unknown.

The system $\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}$ is solvable if and only if the vector $\boldsymbol{y}$ can be expressed as a combination of the columns of $\boldsymbol{A}$.

Then $\boldsymbol{y}$ is in the column space.

figure from Linear Algebra and its applications by Gilbert Strang

The Nullspace of $\boldsymbol{A}$

Nullspace $\boldsymbol{N}(\boldsymbol{A})$

The nullspace of a matrix consists of all vectors $\boldsymbol{x}$ such that $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{0}$.

$$\begin{bmatrix} 1 & 0 \\ 5 & 4 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Nullspace contains $(0,0)$.

$$\begin{bmatrix} 1 & 0 & 1 \\ 5 & 4 & 9 \\ 2 & 4 & 6 \end{bmatrix} \begin{bmatrix} c \\ c \\ -c \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Nullspace is a line.

Linear Independence

Linear Independence: Given a set of vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p$, suppose

$$c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 + \cdots + c_p\boldsymbol{v}_p = \mathbf{0}$$

Only happens when $c_1 = \cdots = c_p = 0$.

$$\boldsymbol{A} = \begin{bmatrix} | & | & & | \\ \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_p \\ | & | & & | \end{bmatrix}$$

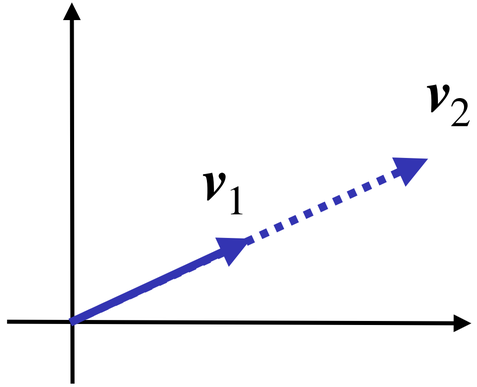
Then, the vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p$ are **linearly independent**.

$\boldsymbol{N}(\boldsymbol{A}) = [\textit{zero vector}]$

Otherwise, the $\boldsymbol{v}$'s are linearly **dependent,**
i.e. one vector is a combination of the others.

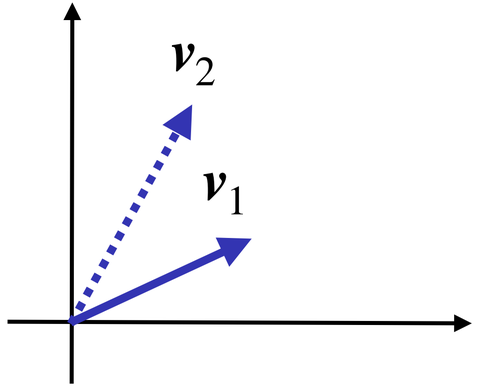
Linear Independence/Dependence Example

Linear Independence: Neither of the vectors can be represented as linear combination of the others.

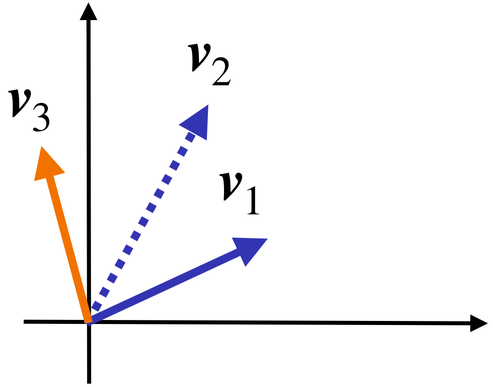


$\boldsymbol{v}_2 = 2\boldsymbol{v}_1$ *Linearly dependent*

$2\boldsymbol{v}_1 - \boldsymbol{v}_2 = 0$



Linearly independent



Linearly dependent

To cover the $\mathbb{R}^n$ space, at least how many vectors you need to represent it?

1. Go to wooclap.com
2. Enter the event code in the top banner

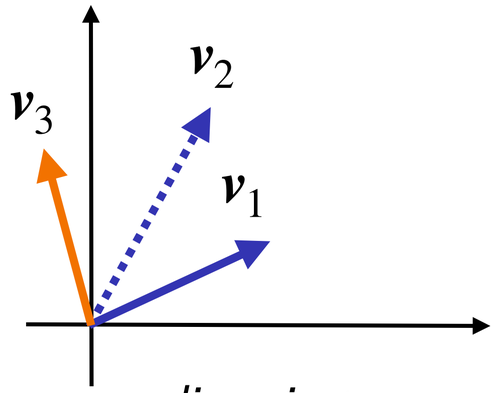
Event code
BDMDUZ

Enable answers by SMS

How Many Vectors to Represent an n-dimensional Space?

To cover the $\mathbb{R}^n$ space, you need vectors whose linear combinations cover the whole space.

e.g. n independent vectors.



$\boldsymbol{v}_i$ lives in two-dimensional space.

Linearly dependent

A set of p vectors in $\mathbb{R}^n$ be must linearly dependent if $p > n$.

Matrix Property to Check Vectors' Independence

Given $\boldsymbol{v}_1, \boldsymbol{v}_2, \ldots, \boldsymbol{v}_p$, construct

$$\boldsymbol{A} = \begin{bmatrix} | & | & & | \\ \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_p \\ | & | & & | \end{bmatrix} \in \mathbb{R}^{n \times p}$$

Then, solve the system $\boldsymbol{A}\boldsymbol{c} = \boldsymbol{0}$.

The vectors are **dependent** if there is a solution other than $\boldsymbol{c} = \boldsymbol{0}$.

Rank of a Matrix

Rank of $\boldsymbol{A}$

The max number of linearly independent columns (or rows) in matrix $\boldsymbol{A} \in \mathbb{R}^{n \times p}$.

$$\mathrm{Rank}(\boldsymbol{A}) := r \qquad \begin{array}{l} r \leq n \\ r \leq p \end{array}$$

Full Rank

A matrix is full rank when its rank is as large as possible, i.e., $r = min(n, p)$.

Vector Space and Subspace

Vector Space

Examples are $\mathbb{R}, \mathbb{R}^2, \mathbb{R}^3, \ldots$; the space $\mathbb{R}^n$ consists of all column vectors with n components. Within all vector spaces, we can take linear combination of vectors.

Subspace

Subspace is a nonempty subset that satisfies the requirements for a vector space: ***Linear combinations stay in the subspace****.*

(i) If we add any vectors $\boldsymbol{x}$ and $\boldsymbol{y}$ in the subspace, $\boldsymbol{x}+\boldsymbol{y}$ is in the subspace.
(ii) If we multiply any vector $\boldsymbol{x}$ in the subspace by any scalar c, $c\boldsymbol{x}$ is in the subspace.

Spanning a Subspace

If a vector space $\boldsymbol{V}$ consists of all linear combinations of $w_1, w_2, \ldots, w_l$, then these vectors span the space.

Every $\boldsymbol{v}$ comes from $\boldsymbol{w}$'s

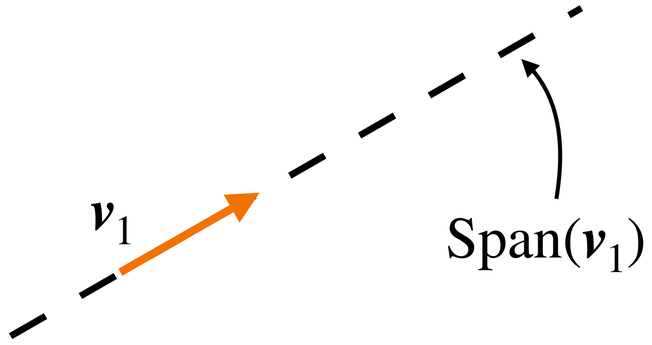
$$\boldsymbol{v} = c_1\boldsymbol{w}_1 + \cdots + c_l\boldsymbol{w}_l$$

for some coefficients c_i.

$\boldsymbol{w}$'s span the vector space $\boldsymbol{V}$.

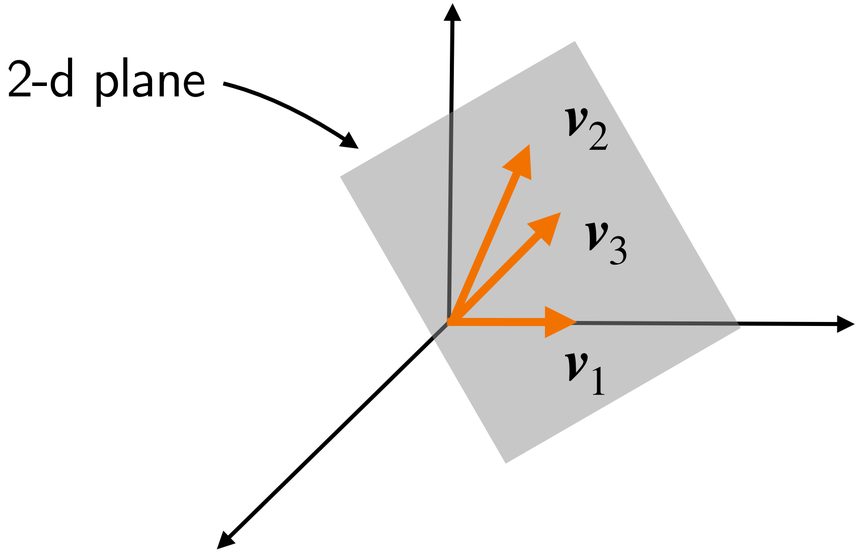
Span Examples

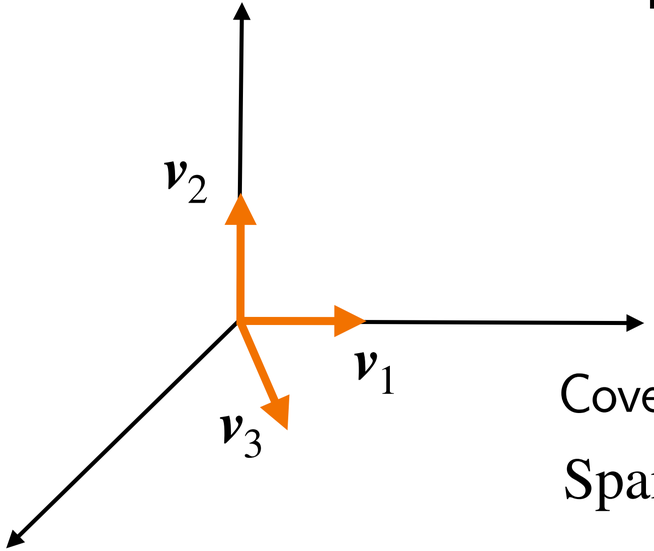
Ex1. $\text{Span}(\boldsymbol{v}_1) = \{c_1\boldsymbol{v}_1 \,|\, c_1 \in \mathbb{R}\}$



Ex2. $\text{Span}(\boldsymbol{A}) = \{c_1\boldsymbol{v}_1 + c_2\boldsymbol{v}_2 + c_3\boldsymbol{v}_3 \,|\, c_1, c_2, c_3 \in \mathbb{R}\}$

$$\boldsymbol{A} = \begin{bmatrix} | & | & | \\ \boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{v}_3 \\ | & | & | \end{bmatrix} \in \mathbb{R}^{3\times 3}$$





Cover 3-d space

$\text{Span}(\boldsymbol{A}) = \mathbb{R}^3$

Basis for a Vector Space

A **basis** for vector space $\boldsymbol{V}$ is a sequence of vectors that

I) *The vectors are linearly independent (not too many vectors).*

II) *They span the space* $\boldsymbol{V}$ *(not too few vectors).*

Remark.

There is **one and only one** way to write $\boldsymbol{v}$ as a combination of the basis vector.

Coordinate Vectors

These columns $\boldsymbol{e}_1$, $\boldsymbol{e}_2$, $\boldsymbol{e}_3$, and $\boldsymbol{e}_4$ represent unit vectors in the coordinate directions in $\mathbb{R}^4$

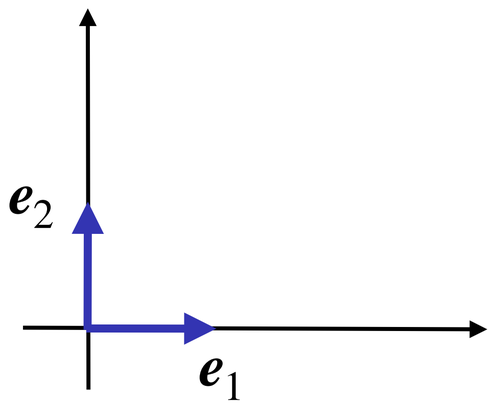
$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad e_4 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}.$$

This can be extended to $\mathbb{R}^n$.

Example of Basis

The coordinate vectors $e_1, \ldots, e_n$ are a basis for $\mathbb{R}^n$.

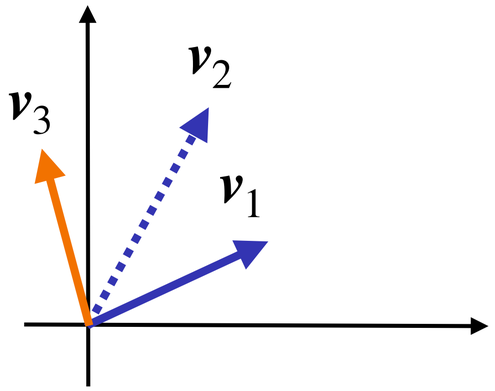
Note: This is not unique. You can define infinitely many different bases.



Number of the basis vectors

Dimension of the vector space

Any two of these vectors, say $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ span $\mathbb{R}^2$, and they are independent. Hence, they form a *basis*.



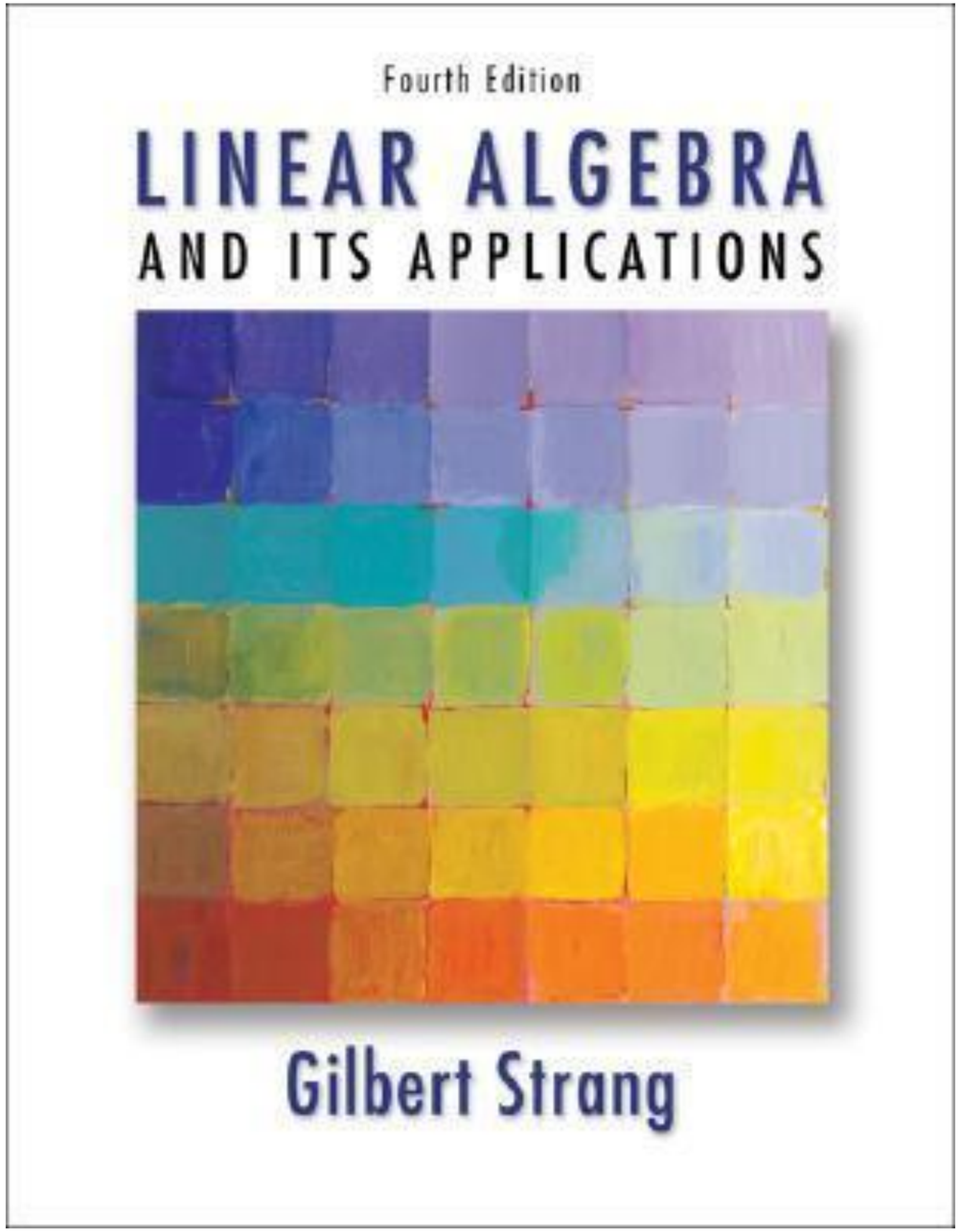
Basis, Independent Sets, and Spanning Sets

I) In a subspace of dimension k, no set of more than k vectors can be independent, and no set of fewer than k vectors can span the space.

II) A basis is a **maximal independent set**. It cannot be made larger without losing independence.

III) A basis is a also **minimal spanning set**. It cannot be made smaller and still span the space.

Wanna be A Machine Learning Expert?

Learn *Probability*, ***Linear Algebra***, and *Optimization* deeply.

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